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Access Bus

The diagram illustrates the Access Bus connections for the HN1x8 module. It shows a set of 8 green horizontal lines representing the bus. On the left, these lines are labeled with the following signals: ESP_EN, GPIO2_U1RXD, GPIO34_U1RXD, GPIO0_U1RTS, and GPIO39_U1CTS. On the right, the lines are numbered 1 through 8, with the label 'Access_Bus1' above them. Below the numbers, there is a vertical stack of 8 pins. To the left of the bus lines, there are three labels: +5V, +3.3V, and GND, each with a red arrow pointing to the corresponding line (lines 1, 2, and 3 respectively).

Signal	Line Number
+5V	1
+3.3V	2
GND	3
ESP_EN	4
GPIO2_U1RXD	5
GPIO34_U1RXD	6
GPIO0_U1RTS	7
GPIO39_U1CTS	8

HN1x8

USB - UART1

+5V_USB

A4
B9
B4
A9
VBUS

DN1 A7
DN2 B7
DP1 A6
DP2 B6
CC1 A5
CC2 B5
SB1 A8
SB2 B8
GND
B1
B12
A12
SHELL 0

GND

U3

I/O1 1
I/O2 2
I/O3 3
I/O4 4
I/O5 5
I/O6 6
I/O7 7
I/O8 8
I/O9 9
I/O10 10
I/O11 11
I/O12 12
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VGA

The diagram illustrates the pinout for a VGA connector, showing 17 pins and their connections to various components. The pins are numbered 1 through 17, with pins 1 through 16 being color-coded (Red, Green, Blue, Yellow, Red, Blue, Red, Blue, Key, GND, ID0, ID1, ID2, ID3, ID4, ID5, ID6, ID7, ID8, ID9, ID10, ID11, ID12, ID13, ID14, ID15, ID16, ID17). The connections are as follows:

- Pin 1 (Red):** Red
- Pin 2 (Green):** Green
- Pin 3 (Blue):** Blue
- Pin 4 (Yellow):** ID2/RES
- Pin 5 (Red):** GND
- Pin 6 (Blue):** Red_RTN
- Pin 7 (Red):** Blue_RTN
- Pin 8 (Blue):** Red_RTN
- Pin 9 (Yellow):** Key/PWR
- Pin 10 (Red):** GND
- Pin 11 (Blue):** ID0/RES
- Pin 12 (Green):** ID1/SDA
- Pin 13 (Blue):** HSync
- Pin 14 (Yellow):** VSync
- Pin 15 (Red):** ID3/SC/L
- Pin 16 (Blue):** Shield1
- Pin 17 (Yellow):** Shield2

The connections to various components are as follows:

- GPIO23/Red_1/LCD_DC** (Pin 1)
- GPIO21/Red_0/LCD_RST** (Pin 2)
- GPIO19/Green_1/VSP/MISO** (Pin 3)
- GPIO18/Green_0/VSP/SCK** (Pin 4)
- GPIO5/Blue_1/VSP/CS** (Pin 5)
- GPIO4/Blue_0/V2C_SDA** (Pin 6)
- GPIO23/HSync/VSP/MOSI** (Pin 13)
- GPIO15/VSync/V2C_SCL** (Pin 14)

The diagram also shows connections to various components like **GPIO23/Red_1/LCD_DC**, **GPIO21/Red_0/LCD_RST**, **GPIO19/Green_1/VSP/MISO**, **GPIO18/Green_0/VSP/SCK**, **GPIO5/Blue_1/VSP/CS**, **GPIO4/Blue_0/V2C_SDA**, **GPIO23/HSync/VSP/MOSI**, and **GPIO15/VSync/V2C_SCL**.

HDR15-3-0B-14-51/VGA615/14.5mm

The circuit diagram shows the internal components of the Audio module. It includes two 3.5mm audio jacks labeled AUDIO1 and BUZZER. The BUZZER jack is connected to a buzzer component. A 22µF electrolytic capacitor (C19) is connected between the BUZZER jack and ground. A 270Ω resistor (R41) is connected between the BUZZER jack and the GPIO25/AUDIO pin. A 600Hz/10V/1K/X7R/C0402 ceramic capacitor (C20) is connected between the BUZZER jack and ground. A 1MΩ/4W/500I23 resistor (D5) is connected between the BUZZER jack and ground. A 1N4148W/SOD123 diode (D6) is connected between the BUZZER jack and ground. A 3.3V voltage regulator (BUZ_EN1) is connected between the BUZZER jack and ground. A 3.3V voltage regulator (BUZ_EN1) is connected between the BUZZER jack and ground.

LCD Connector

The diagram illustrates the pinout for an LCD connector. It consists of 12 pins, numbered 1 through 12. The functions and power levels for each pin are as follows:

Pin Number	Function	Power Level
1	GPI05\Blue_1\VSPL_CS	+3.3V
2	GPI022\Red_1\LCD_DC	GND
3	GPI018\Green_0\VSPL_SCK	
4	GPI023\HSync\VSPL_MISO	
5	GPI019\Green_1\VSPL_MISO	
6	GPI021\Red_0\LCD_RST	
7	GPI015\VSync\I2C_SCL	
8	GPI04\Blue_0\I2C_SDA	
9		
10		
11	GND	
12	+5V	

The connector is labeled **LCD_CON1** at the top right and **HN1x12** at the bottom right.

Software Selectable Pins		
Interface	Signal	Pin
EMAC	EMAC_MDC_out	Any GPIO
	EMAC_MDI_in	
	EMAC_MDIO_out	
	EMAC_CRDS_out	
	EMAC_CDL_out	
I2C	I2CEXT0_SCL_in	Any GPIO
	I2CEXT0_SDA_in	
	I2CEXT1_SCL_in	
	I2CEXT1_SDA_in	
	I2CEXT0_SCL_out	
	I2CEXT0_SDA_out	
	I2CEXT1_SCL_out	
General Purpose SPI	HSPIQ_in/_out	Any GPIO
	HSPI0_in/_out	
	HSPICLK_in/_out	
	HSPLCS0_in/_out	
	HSPLCS1_out	
	HSPLCS2_out	
	VSPIQ_in/_out	
	VSPI0_in/_out	
	VSPICLK_in/_out	
	VSPLCS0_in/_out	
	VSPLCS1_out	
	VSPLCS2_out	

For more information refer to [esp_wroom_32_datasheet_en.pdf](#).

PS/2 Port 0 -> Keyboard

Current Consumption Approximately: 50mA

PS2_KEYBOARD1

IO	1	1	PS2_DAT
NA	2	2	+
GN	3	3	+
SV	4	4	+
CK	5	5	PS2_CLK
NA	6	6	+
NC1	7	7	+
NC2	8	8	+
SHIELD1	9	9	+

MDR6_MINI-DIN

GND GND

5V

R19 10k/R0402

R28 120R/R0402

GPI033\KBD_DAT

5V

R20 10k/R0402

R29 120R/R0402

GPI033\KBD_CLK

100nF/50V/0805/0402

[illegible]

Mounting Holes



MH1

Mounting_Hole_NPTH



MH2

Mounting_Hole_NPTH



MH3

Mounting_Hole_NPTH



MH4

Mounting_Hole_NPTH

Fiducials

 FID1 Fiducial	 FID3 Fiducial	 FID5 Fiducial
 FID2 Fiducial	 FID4 Fiducial	 FID6 Fiducial

[illegible]

pander

UEXT

UEXT_Pwr_EN# to be included by default!

PC3\UEXT_PWR_EN#

R30 2.2k/0402

R31 33k

220V/0402

C17 22uF/25V/0402

25V/6.3V/20V/0402

FET2 WPM2015-3/TR

+3.3V

R32 2.2k/0402

R37 2.2k/0402

UEXT1

1 2 3 4 5 6 7 8 9 10

B-R-10-LF (0H10R)

PD5\UTX
PC3\SCI
PA2\SoftSPL_MISO
PD4\SoftSPL_SCK

PD6\VRX
PC3\SDA
PA1\SoftSPL_MOSI
PD3\SoftSPL_CS#